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KAMELIA CHEKROUD

PhD Student in Artificial Intelligence and Software Engineering

Contact

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Language

- **Arabic :** Proficient (C1 Level – Hamza Certificate, score: 100%)
- **English :** Intermediate (B1 Level – EF SET Certificate, score: 48%)
- **French :** Intermediate / Good

Soft Skills

- Communication skills
- Teamwork
- Responsibility
- Adaptability
- Problem solving
- Motivation to learn
- Discipline

About Me

I am a PhD student in Artificial Intelligence and Software Engineering, with practical experience in full-stack web development, database management, and applied machine learning. My research interests focus on human-machine collaboration, intelligent systems, and the development of AI-based solutions for real-world challenges.

Motivated, adaptable, and research-oriented, I am eager to further develop my academic and technical skills through international learning and research opportunities. My goal is to contribute to innovative projects that combine artificial intelligence, software engineering, and human-centered technologies, while strengthening my expertise through academic collaboration and advanced training.

Education

PhD in Artificial Intelligence and Software Engineering

National Higher School of Advanced Technologies — ENSTA, Algiers, Algeria - 2025

Thesis Title: Human-Machine Collaborative Control of Industry 4.0 Assembly Systems

Research focus:

Human-machine collaboration, intelligent control systems, interactive decision-making, optimization, and artificial intelligence for advanced manufacturing systems.

Master's Degree in Software Engineering and Information Processing

Faculty of Sciences — University M'Hamed Bougara of Boumerdès, Algeria

2022 – 2024

Main areas of study:

Software engineering, information systems, data processing, artificial intelligence, web development, and database systems.

Bachelor's Degree in Mathematics and Computer Science

Specialization: Information Systems and Software Engineering

Faculty of Sciences — University M'Hamed Bougara of Boumerdès, Algeria

2019 – 2021

Baccalaureate in Experimental Sciences

Algerian Baccalaureate in Experimental Sciences - 2017, with a Pass distinction

High School, Algeria

Technical skills

- **Programming Languages:** Python, JavaScript, PHP, Java, C/C++
- **Artificial Intelligence & Data Science:** Deep Learning, Convolutional Neural Networks, VQ-VAE, K-Means, PyTorch, NumPy, SciPy
- **Web Development:** React, HTML, CSS, Bootstrap, Tailwind CSS, Flask, Django, Node.js, PHP
- **Database Management:** MySQL, PostgreSQL, SQLite, SQLAlchemy
- **Software Engineering:** UML Modeling, Object-Oriented Programming, Full-Stack Development, CRUD Applications
- **Tools & Technologies:** Git, VS Code, XAMPP, LaTeX, Jinja2, Flask-Babel
- **Research & Academic Skills:** Scientific research, Academic and Report Writing, Literature Review, Research Methodology, Data organization and Analysis, Academic Presentation, Teaching Assistance, Practical Lab Supervision, Self-learning, Time management

Academic and Professional Experience

Temporary Lecturer

Department of Computer Science, Faculty of Sciences, University M'Hamed Bougara of Boumerdès
2024 – 2025
Delivered tutorials and practical sessions in computer science subjects, including:
Machine Structure
OPM
MATLAB
Office Automation
Web Development
Responsibilities included preparing practical sessions, assisting students during laboratory work, explaining technical concepts, and supporting students in programming and software-related assignments.

Final Year Research Project — Medical Data Compression Using Deep Learning

Department of Computer Science, Faculty of Sciences, University M'Hamed Bougara of Boumerdès
2023 – 2024
Worked on the design and implementation of an efficient medical dataset compression approach using deep learning models. The main objective was to reduce the size of medical datasets while preserving their diagnostic relevance.
Key aspects:
Developed a deep learning-based compression method.
Explored neural network architectures for medical data representation.
Focused on maintaining diagnostic utility after compression.
Used machine learning tools and scientific computing libraries.
Technologies: Python, PyTorch, CNN, VQ-VAE, NumPy, SciPy.

Practical Internship — Web Application for Leave Management

IAP Sonatrach, Boumerdès, Algeria
2021 – 2022
Developed a web application for managing employee leave requests within the Exploration Division of the National Petroleum Institute.
Main tasks:
Designed and implemented a complete leave management system.
Developed authentication and user management features.
Implemented CRUD operations.
Created a structured database for employee and leave records.
Technologies: React, PHP, MySQL.